



Property Valuation Report

Comparable Sales Analysis & Investment Recommendations

Prepared for: CCLBA - Cook County Land Bank Authority
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Document: Automated Valuation Report
Subject: 981 Tax-Delinquent Properties

This report documents the methodology used to estimate market values for Cook County tax-delinquent properties. Each property was enriched using the ATTOM Data API, analyzed through a comparable sales valuation engine, and assigned an investment recommendation based on a \$300,000 threshold.

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HOW WE ESTIMATED THE VALUE OF YOUR PROPERTIES

1. What You Gave Us (Input CSV)

You provided a CSV file named `***CCLBA data 04.2026.csv***` that contained a list of properties.

What was in the file:

Column	What It Contains	Example
Parcel ID	County's unique ID for each property	<code>`31042000050000`</code>
Address	Full street address of the property	<code>`438 N AVERS AVE, CHICAGO, IL 60624`</code>
Owners	Owner name	(this field was blank in the source data)

We processed ****981 properties**** from this list.

2. Where We Got Property Data (ATTOM API)

We used a service called ****ATTOM Data**** (the largest property data provider in the US) to look up each property.

Step 1: Find the Property

We sent ATTOM the address of each property. For example:

`***What we sent:** `438 N AVERS AVE, CHICAGO, IL 60624``

ATTOM searched its database and returned details about that property.

Step 2: What ATTOM Sent Back

For each property, ATTOM returned the following information:

Information We Got	Example Value	Why We Need It
Square footage (Sqft)	1,200 sqft	To compare size with similar properties
Lot size	5,000 sqft	Property land area
Bedrooms	3	Property type info
Bathrooms	1.5	Property type info
Year built	1954	Age of property
Property type	Single Family	House, apartment, etc.
Assessed value	\$12,000	County's tax assessment
Market value	\$180,000	ATTOM's own estimate
Last sale price		

	\$145,000	When it last sold
Last sale date	March 2020	When it last sold
Latitude & Longitude	41.883, -87.724	Location coordinates
ATTOM ID	12345678	ATTOM's internal ID

If ATTOM couldn't find a property (address not in their database), we marked it as "Failed" and moved to the next one. This happens for about 1-2% of addresses.

3. Where We Got Comparable Sales (ATTOM Comps API)

What Are Comparable Sales?

Comparable sales (or "comps") are **recently sold properties that are similar to yours and located nearby**. Real estate agents use the same concept when pricing a house.

How We Found Comps

We used two methods:

Method A (Preferred): If ATTOM gave us an ATTOM ID for the property, we used their special "Sales Comparables" tool that automatically finds the most relevant nearby sold properties.

Method B (Fallback): If we only had coordinates (latitude/longitude), we searched for all sold properties within a **1-mile radius** that sold in the **last 12 months**.

What Data We Got for Each Comparable Sale

Information	Example	Used For
Sale price	\$200,000	Main input for valuation
Sale date	2025-08-15	To check if sale is recent
Square footage	1,250 sqft	For price-per-sqft calculation
Address	440 N AVERS AVE	Just for identification

We **skipped** any comps with a \$0 sale price (these are usually tax transfers, not real market sales).

Typical Results

- Properties with good comps: **5-10** nearby sold properties found
- Some properties: **0** comps (too rural, unique property type)
- Average comps per property: **4-6**

4. How We Cleaned the Data First

Before running the valuation, we fixed some issues in your CSV:

Problem 1: Excel Format in Parcel IDs

Your CSV had parcel IDs like `"=31042000050000"` (Excel formula format). We stripped it to just `31042000050000`.

Problem 2: Duplicate Properties

About **30-40** properties appeared twice in the list. We removed duplicates.

Problem 3: Address Format

ATTOM reads addresses better in UPPERCASE, so we converted everything to capital letters and removed special CCLBA annotations like "HSE" (meaning "house") that confused ATTOM.

5. How We Removed Bad Comparable Sales (Outlier Removal)

Before calculating value, we **remove extreme prices** from our comparable sales list. This is important because:

Real Example

Imagine we find 10 nearby sold properties with these prices:

\$180,000	\$185,000	\$190,000	\$195,000	\$200,000
\$205,000	\$210,000	\$215,000	\$220,000	\$2,000,000

The last property at **\$2 million** is clearly different (maybe a mansion while others are small homes). If we include it, the average would be falsely high.

The Formula We Used (IQR Method)

We used a statistical method called **IQR (Interquartile Range)** to automatically detect and remove such outliers:

```

Step 1: Sort all prices from lowest to highest
Step 2: Find Q1 = the price at the 25% mark
Step 3: Find Q3 = the price at the 75% mark
Step 4: IQR = Q3 - Q1 (the middle spread)
Step 5: Lower limit = Q1 - (1.5 x IQR)
Step 6: Upper limit = Q3 + (1.5 x IQR)
Step 7: Remove any price below lower limit or above upper limit

```

Simple Analogy

Think of it like judging a diving competition. The judges drop the highest and lowest scores before calculating the average. We do the same -- any comp that is **way too high or way too low** compared to the rest is removed.

Safety check: If removing outliers would remove ALL comps (very rare), we keep the original list.

6. Our Valuation Formula -- Step by Step

After removing bad comps, we use **two different methods** to estimate the property value, then combine them.

Method A: Median Price (Simple & Reliable)

We take the **middle value** (median) of all comparable sale prices.

Formula:

Median Estimate = The middle price when all comp prices are sorted

Example: If comp prices are: \$180K, \$190K, **\$200K**, \$210K, \$220K

- The median is **\$200,000** (the middle value)

Why median, not average?

- Average (mean) = all prices added divided by count
- If one comp is \$2M instead of \$200K, the average jumps to \$560K -- wrong!
- The median stays at **\$200K** no matter what -- it is stable

Method B: Price-per-Square-Foot (Size Adjustment)

This method adjusts for size differences between your property and the comparable sales.

Step-by-step:

Step 1: For each comp, calculate:

$$\text{Price per sqft} = \text{Sale Price} / \text{Square Footage}$$

Step 2: Take the average of all price-per-sqft values

(also remove outliers here using the same IQR method)

Step 3: Multiply by your property's square footage:

$$\text{Sqft Estimate} = \text{Average Price per sqft} \times \text{Your Property's Sqft}$$

Real Example:

Comp	Sold For	Size	Price per Sqft
House A	\$200,000	1,200 sqft	\$166.67/sqft
House B	\$220,000	1,300 sqft	\$169.23/sqft
House C	\$190,000	1,150 sqft	\$165.22/sqft

Average price per sqft = $(\$166.67 + \$169.23 + \$165.22) / 3 = \$167.04/\text{sqft}$

If your property is **1,250 sqft**:

$$\text{Sqft Estimate} = \$167.04 \times 1,250 = \$208,800$$

Final Blend: Combining Both Methods

We don't rely on just one method. We combine them with **60% weight on the Median** and **40% weight on the Sqft method**:

$$\text{Final Value} = (\text{Median} \times 0.6) + (\text{Sqft Estimate} \times 0.4)$$

Why 60/40?

- The Median is more **stable** (outliers can't affect it) -- so it gets 60%
- The Sqft method adds **size adjustment** -- it gets 40%
- If we don't know your property's square footage, we use only the Median

Example calculation:

Median Estimate = \$200,000

Sqft Estimate = \$208,800

$$\begin{aligned} \text{Final Value} &= (\$200,000 \times 0.6) + (\$208,800 \times 0.4) \\ &= \$120,000 + \$83,520 \\ &= \$203,520 \end{aligned}$$

7. How Confident Are We in Each Estimate?

Not all estimates are equally reliable. We assign a **confidence level** based on two factors:

Factor 1: How Many Comps Did We Find?

- **3+ comps** = Good (more data = more reliable)
- **1-2 comps** = OK
- **0 comps** = Cannot estimate

Factor 2: How Consistent Are the Comps? (Variance)

We calculate **variance** -- a measure of how spread out the comp prices are:

$$\text{Variance \%} = (\text{Standard Deviation of comp prices} / \text{Estimated Value}) \times 100$$

Plain English: What Does Variance Mean?

If comps look like this	Variance	Meaning
\$190K, \$195K, \$200K, \$205K, \$210K	**Low (~4%）**	Comps are very consistent -- high reliability
\$150K, \$200K, \$250K, \$300K, \$350K	**High (~38%）**	Comps are all over the place -- low reliability

Confidence Levels

Confidence	When Does It Happen?	What It Means
HIGH	3+ comps AND prices are consistent (variance under 20%)	You can rely on this estimate
MEDIUM	1-2 comps OR prices are somewhat spread out (variance 20-40%)	Reasonable estimate, but some uncertainty
LOW	Very few comps OR prices are highly scattered (variance over 40%)	Use with caution
NONE	No comps found at all	We cannot estimate

8. Investment Recommendation (YES / NO / MAYBE)

Based on the estimated value, we flag each property:

The Threshold: \$300,000

We set **\$300,000** as the main decision line. But we don't use a hard cut -- we add a **25% margin** to avoid borderline mistakes.

The Three Zones

SKIP ZONE	REVIEW ZONE	BUY ZONE
????????????????	????????????????	????????????????
	\$225,000	\$375,000

Estimated Value	Flag	What To Do
Above \$375,000	**YES**	This property is likely worth significantly more than \$300K -- worth pursuing
Below \$225,000	**NO**	This property is well below \$300K -- likely not worth it
Between \$225K and \$375K	**MAYBE**	Borderline case -- needs manual review
No estimate	**UNKNOWN** *	Not enough data to evaluate

Why a Margin Zone?

A property worth \$299,000 is essentially the same as one worth \$301,000. If we used a strict "\$300K or above" rule, one would be NO and the other YES -- which makes no sense. The 25% margin (\$75,000 buffer) catches these borderline cases and flags them for **human review**.

9. The Output File You Receive

After processing, we save the results as a CSV file in the `output` folder:

```
output/results_[job_id]_[date]_[time].csv
```

What Each Row Contains

Column	Description
Parcel ID	County's unique ID
Address	Full address of the property
Investment Flag	YES / NO / MAYBE / UNKNOWN
Estimated Value	Our calculated market value
Confidence	HIGH / MEDIUM / LOW / NONE
Comp Count	Number of comparable sales used (after removing bad ones)
Price/sqft	Average price per square foot from comps
Sqft	Property's square footage
Beds	Number of bedrooms
Baths	Number of bathrooms
Year Built	Year the property was built
Assessed Value	County's assessment
Last Sale Price	Most recent sale price
Last Sale Date	When it last sold
Prop Type	Single Family, Multi-Family, etc.
Status	success / partial / failed
Notes	Processing notes (e.g. "5 comps used, 1 removed, variance 15%")

Typical Notes You'll See

```
5 comp(s) used | 1 outlier(s) removed | variance 15.2% | size-adjusted
```

This means:

- We found 5 comparable sales
- 1 was removed as an outlier (too high or too low)
- The remaining 4 have a variance of 15.2% (consistent)
- We used the size-adjustment method (sqft was available)

10. Summary of Results So Far

Across all 981 properties processed:

Metric	Count
Total properties	
	981
YES (Buy)	
	(varies by run)
NO (Skip)	
	(varies by run)
MAYBE (Review)	
	(varies by run)
Failed (no data)	
	(varies by run)

Appendix: Complete Flow Diagram (Simple Version)

YOUR CSV FILE

```

      |
      v
+-----+
| Step 1: Clean the Data |
| - Fix parcel IDs      |
| - Remove duplicates   |
| - Fix missing ZIPs    |
+-----+
      |
      v
+-----+
| Step 2: ATTOM Lookup  |
| Send address          |
| Get: sqft, beds,     |
| baths, assessed value |
+-----+
      |
      v
+-----+
| Step 3: Find Comps    |
| Search nearby sold    |
| properties (1 mile,   |
| 12 months)            |
+-----+
      |
      v
+-----+
| Step 4: Remove Bad    |
| Comps (Outlier Filter)|
+-----+
      |
      v
+-----+
| Step 5: Calculate     |
| Value                 |
| Median (60%) +        |
| Sqft Method (40%)     |
+-----+
      |
      v
+-----+
| Step 6: Confidence    |
| & Investment Flag     |
+-----+
      |
      v
      YOUR OUTPUT CSV
(Ready to review or share)
```

End of Report